SCHOOL OF NATURAL SCIENCES

DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE

BACHELOR OF SCIENCE HONOURS DEGREE (COMPUTER SCIENCE)

LEVEL 2 SEMESTER1

EXAMINATION QUESTION PAPER

MODULE CODE	HCS225
MODULE NARRATION	DESIGN AND ANALYSIS OF ALGORITHMS
DATE	2023
DURATION	3 HOURS

INSTRUCTION TO CANDIDATES:

1. Each question carries 20 marks.
2. Answer all questions in section A and any three (3) in section B.

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SECTION A

QUESTION 1

a) Explain what is meant by worst-case efficiency, best-case efficiency and average case efficiency? [6]

b) The elements of an n-by-n array A are arranged in ascending rows and columns; formally, A[i][j] < A[i][j0] whenever j < j0 and A[i][j] < A[i0][j] whenever i < i0. Suppose you wish to determine if A contains a particular target element x. Give the most efficient algorithm you can for solving this problem, and compute the best upper bound you can on its asymptotic worst-case running time as a function of n.

Hint: For some algorithms it may be easier to keep track of the number of elements m than the number of rows or columns. [14]

QUESTION 2

a) Define: i) Profiling ii) Time Complexity iii) Space Complexity. [4]
b) State the Greedy Knapsack Problem. [4]
c) Distinguish between Prim's and Kruskal's Spanning tree algorithm and state which one is better with reasons. [4]
d) Draw all possible binary search trees for the identifier set (do, if, stop). [4]
e) Define Chromatic number & Give the state space tree for 4-coloring problem. [4]

SECTION B

QUESTION 3

a) What are the different mathematical notations used for algorithm analysis. [4]
b) Give the algorithm for transpose of a matrix m x n and determine the time complexity of the algorithm by frequency-count method. [8]
c) Discuss the Amortized analysis with an example. [4]
d) What are the advantages and disadvantages of Divide-And-Conquer? [4]

QUESTION 4

a) State fractional knapsack problem. Give an algorithm for fractional knapsack problem using greedy strategy. [5]
b) Find an optimal solution to the fractional knapsack problem for an instance with number of items 7, Capacity of the sack W=15, profit associated with the items (p1,p2,…,p7)= (10,5,15,7,6,18,3) and weight associated with each item (w1,w2,…,w7)= (2,3,5,7,1,4,1). [5]
c) Define NP hard and NP-Complete problems [4]
b) Write short notes on Polynomial time reductions with example. [4]
c) Define class P and class NP. [2]

QUESTION 5

a) Explain Reliability Design Problem with suitable example. [4]

b) Write an algorithm for N-queen's problem. Give time and space complexity for 8-queen's problem. [8]

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c) Give the statement of sum of subsets problem. Find all sum of subsets for n=4, (w1, w2, w3, w4) = (11, 13, 24, 7) and M=31. Draw the portion of the state space tree using fixed-tuple sized approach. [8]

QUESTION 6

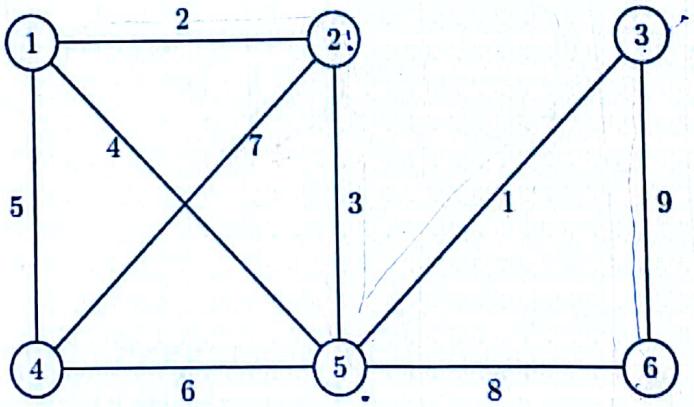
There are n groups of people trying to schedule meetings at the same time, and there are m meeting rooms available. Each group *i (i = 1; 2;... n)* has ai people, and each room j *(j = 1; 2;....m)* has a capacity of bj . Group i can be scheduled to use room j if and only if $ai \leq bj$ (i.e., the number of people is no larger than the capacity of the room).

a) Of these n groups, the first k $(1 \leq k \leq n)$ groups have high priority and they need to be scheduled (there will always be at least one way to schedule them). Design an algorithm that schedules all the high priority group, and schedules as many other groups as possible. [8]

b) Describe your algorithm and analyze its running time. Your algorithm should run in *O(n log n + m log m) time.*

b) Prove the correctness of the algorithm you designed in part (a). [4]

c) Run Prim and Kruskal on the following graph. For Prim's algorithm, start at vertex 1. List the edges in the order that they are added into the minimum spanning tree. Please write an edge as a pair of vertices, like (1,2). Also, please always write the vertex with smaller index in front (that is, write (1,2) instead of (2,1), even though both represent the same edge). [8]



Ordering	1	2	3	4	5
Prim's					
Kruskal					

==========END OF PAPER==========

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